# LSTM results report

Multi-pollutant forecasting | 4 November 2026

This report contains the saved errors and graphs for LSTM.

Dataset: supplied combined hourly CSV with 11 pollutants and 14,592 rows.

Training and validation use 2025. Testing uses January to August 2026.

Daily: hourly forecasts averaged to a day versus a direct daily forecast.

Weekly: daily forecasts averaged to a week versus a direct weekly forecast.

Monthly: weekly blocks averaged to a month versus a direct monthly forecast.

MAE is the average absolute error. RMSE gives larger errors more weight.

Lower MAE and RMSE are better. Compare values within the same pollutant.

Units: CO is mg/m3; NOx is ppb; other pollutants use source micrograms/m3.

Each actual average needs 75% hourly coverage. Missing actuals are not scored.

Forecasts use measurements available before the period starts.

Later 2026 forecasts can use earlier observed 2026 data, without retraining.

Monthly findings are exploratory: only nine direct training windows and eight test months.

One seeded run does not establish superiority. AQI classification is still pending.

Network: one recurrent layer with 32 hidden units; Adam at 0.001; seed 42.

Validate on October-December 2025; select epochs, then refit on eligible 2025 data.

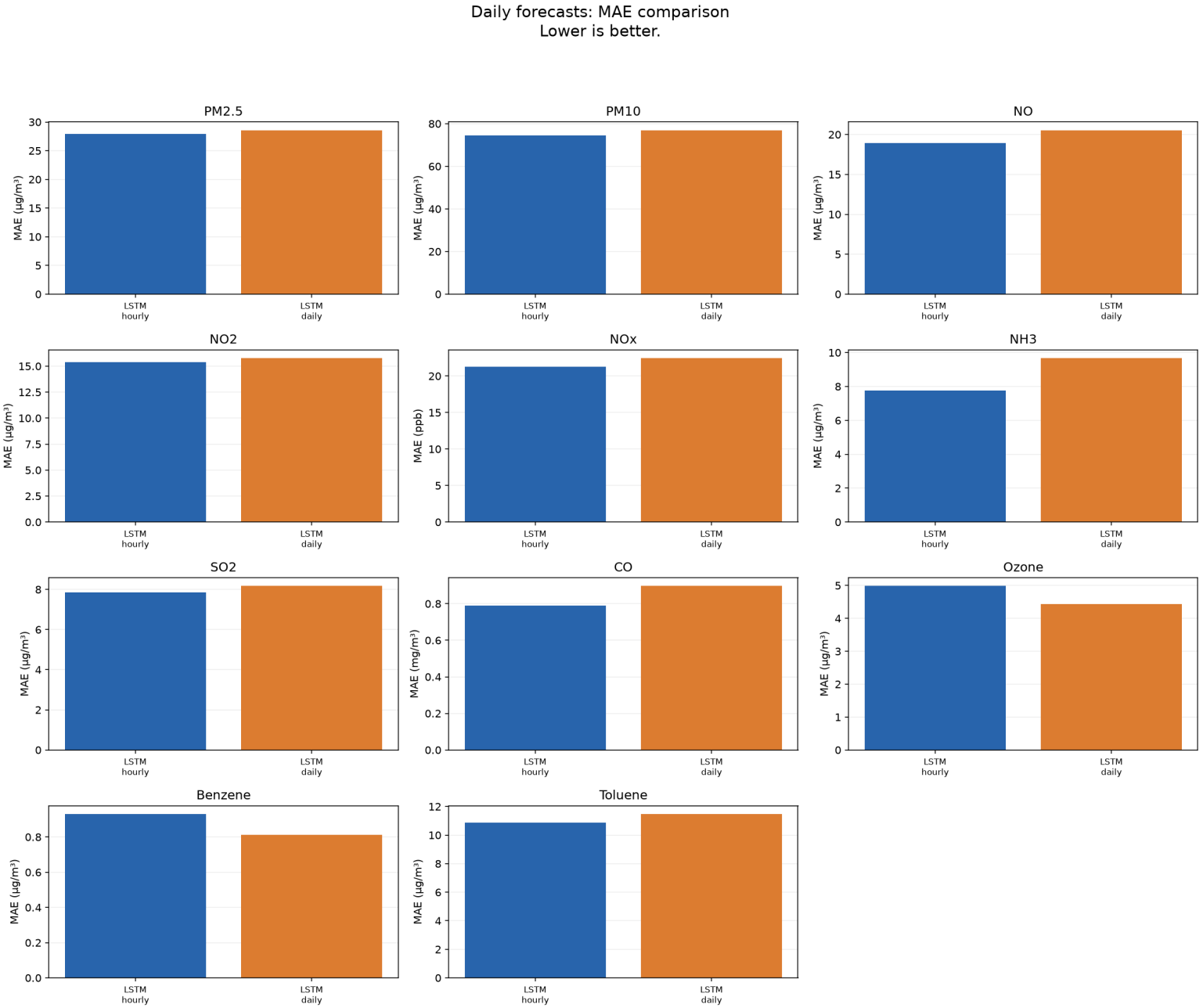
Missing past inputs fill forward or use training medians; scaling uses training data only.

# LSTM daily errors

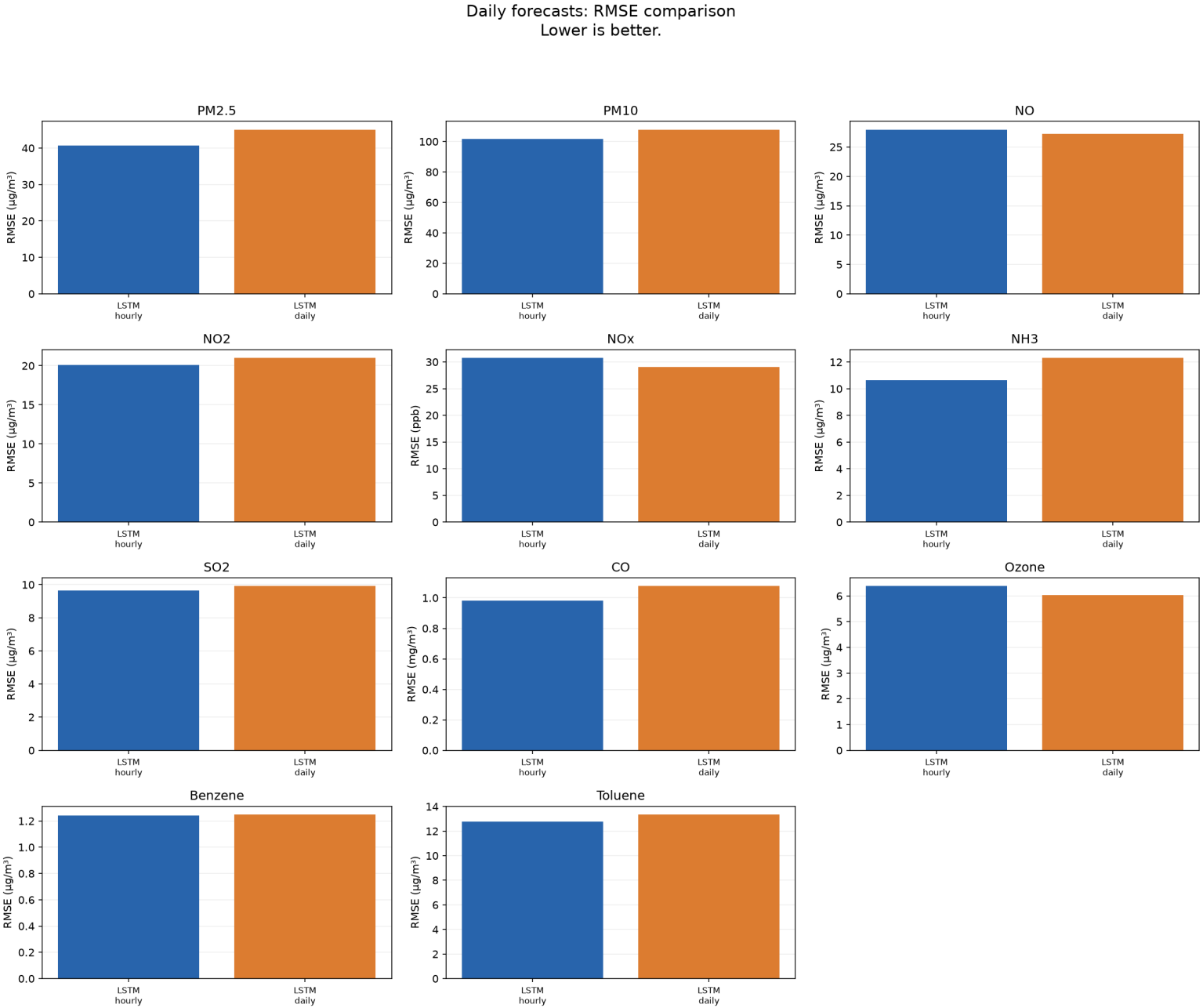
Original concentration units | N is the number of scored periods | Lower is better

| Pollutant | Model | Input scale | N | MAE | RMSE |
|---|---|---|---|---|---|
| Benzene | LSTM | daily | 229 | 0.81 | 1.25 |
| Benzene | LSTM | hourly | 229 | 0.93 | 1.24 |
| CO | LSTM | daily | 228 | 0.90 | 1.08 |
| CO | LSTM | hourly | 228 | 0.79 | 0.98 |
| NH3 | LSTM | daily | 229 | 9.68 | 12.32 |
| NH3 | LSTM | hourly | 229 | 7.76 | 10.65 |
| NO | LSTM | daily | 230 | 20.55 | 27.25 |
| NO | LSTM | hourly | 230 | 18.93 | 27.97 |
| NO2 | LSTM | daily | 227 | 15.78 | 21.01 |
| NO2 | LSTM | hourly | 227 | 15.39 | 20.07 |
| NOx | LSTM | daily | 231 | 22.46 | 29.09 |
| NOx | LSTM | hourly | 231 | 21.27 | 30.76 |
| Ozone | LSTM | daily | 227 | 4.44 | 6.03 |
| Ozone | LSTM | hourly | 227 | 4.98 | 6.38 |
| PM10 | LSTM | daily | 227 | 76.97 | 107.84 |
| PM10 | LSTM | hourly | 227 | 74.46 | 101.61 |
| PM2.5 | LSTM | daily | 228 | 28.58 | 45.09 |
| PM2.5 | LSTM | hourly | 228 | 27.94 | 40.67 |
| SO2 | LSTM | daily | 226 | 8.17 | 9.91 |
| SO2 | LSTM | hourly | 226 | 7.86 | 9.63 |
| Toluene | LSTM | daily | 229 | 11.49 | 13.36 |
| Toluene | LSTM | hourly | 229 | 10.86 | 12.77 |

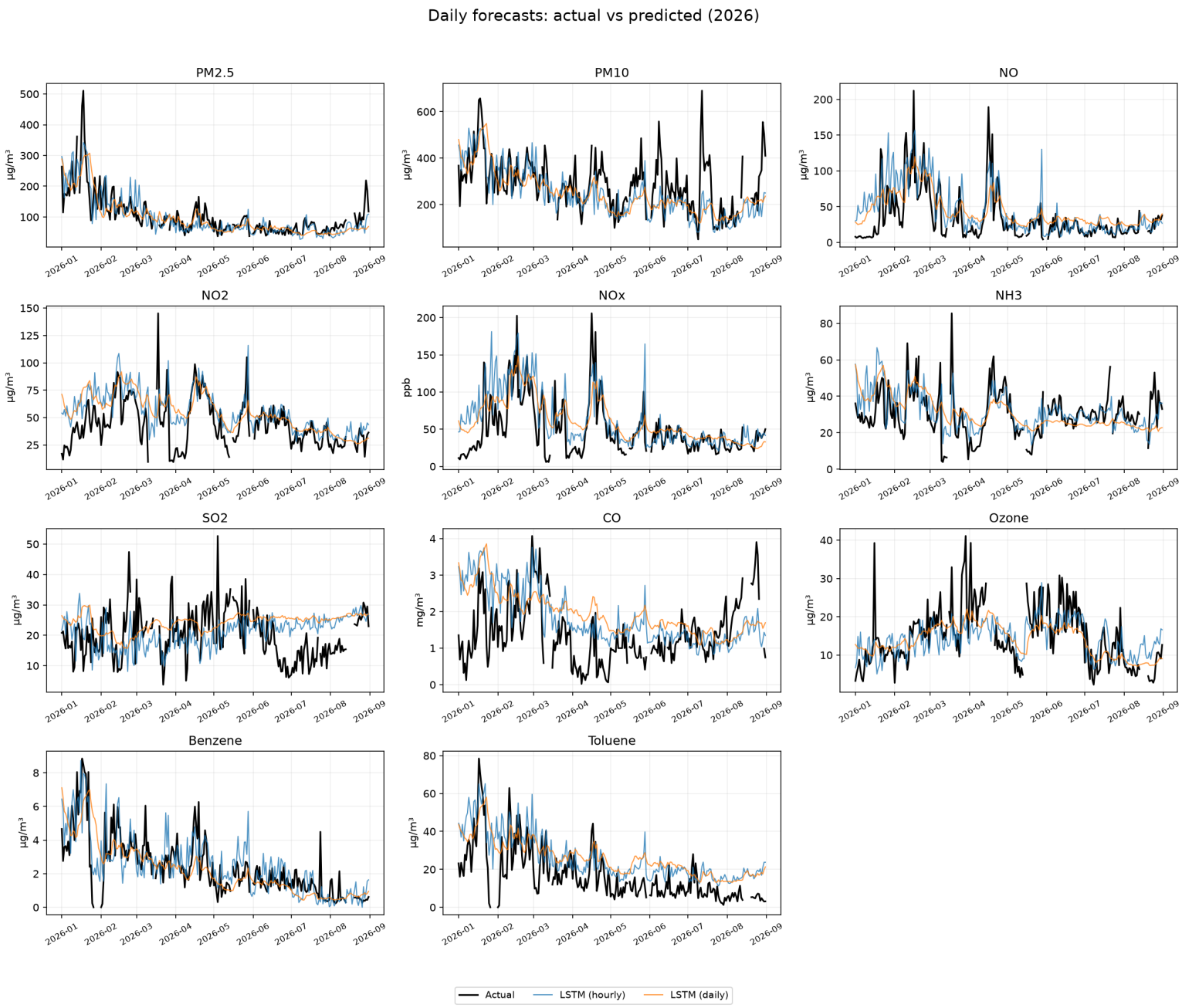
# Daily MAE comparison



# Daily RMSE comparison



# Daily actual and predicted values

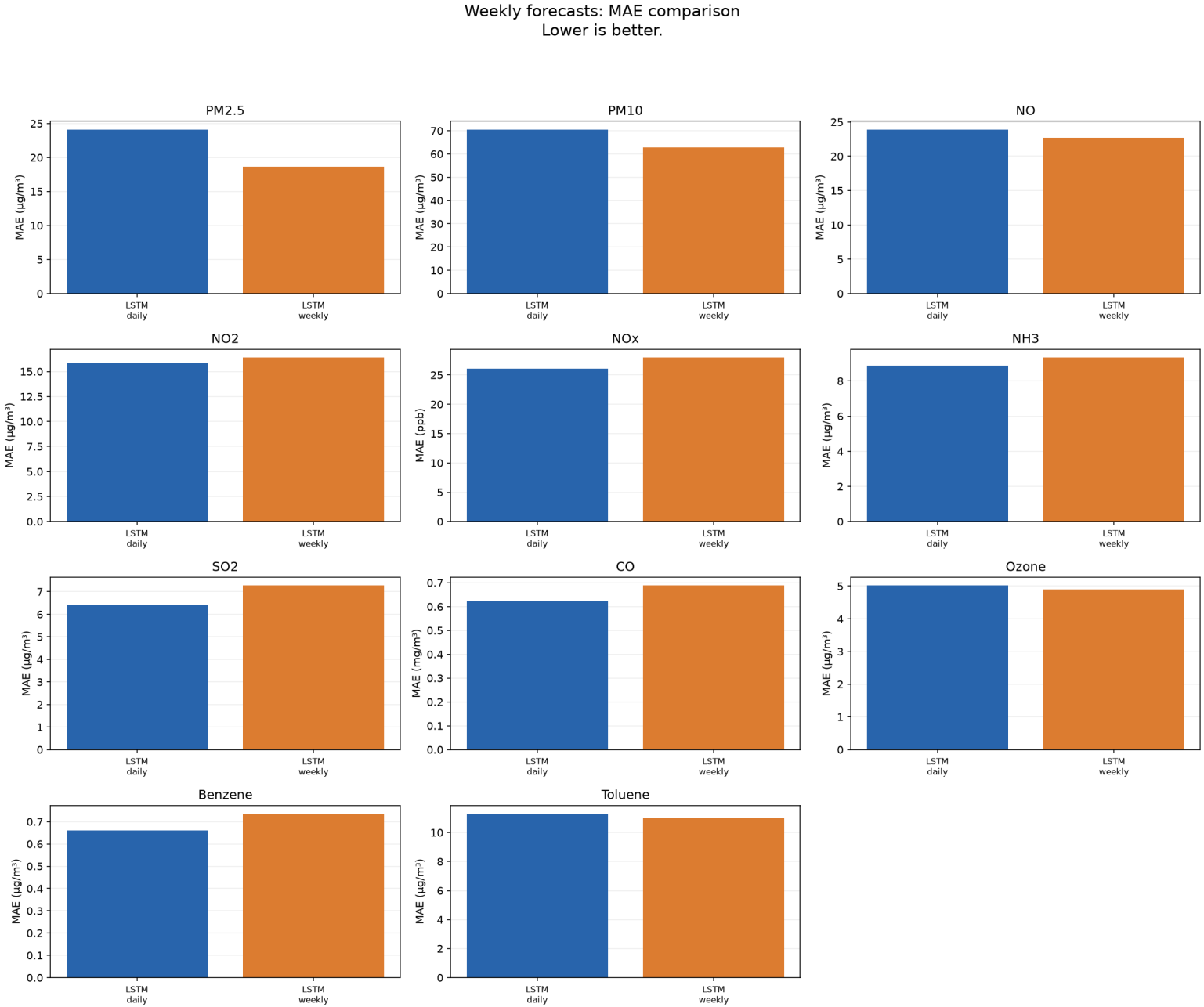


# LSTM weekly errors

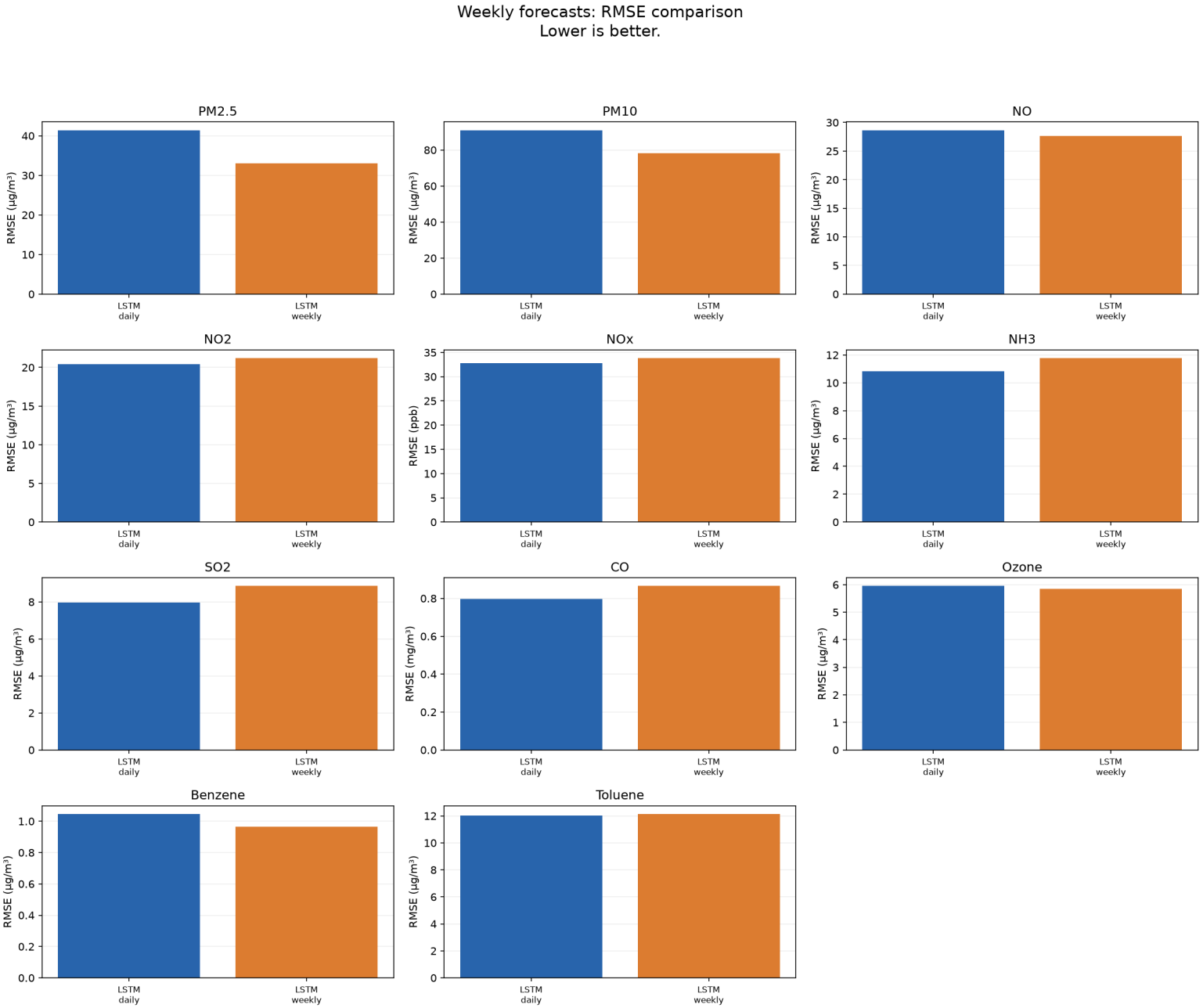
Original concentration units | N is the number of scored periods | Lower is better

| Pollutant | Model | Input scale | N | MAE | RMSE |
|---|---|---|---|---|---|
| Benzene | LSTM | daily | 32 | 0.66 | 1.05 |
| Benzene | LSTM | weekly | 32 | 0.74 | 0.97 |
| CO | LSTM | daily | 32 | 0.62 | 0.80 |
| CO | LSTM | weekly | 32 | 0.69 | 0.87 |
| NH3 | LSTM | daily | 33 | 8.89 | 10.83 |
| NH3 | LSTM | weekly | 33 | 9.34 | 11.78 |
| NO | LSTM | daily | 33 | 23.91 | 28.66 |
| NO | LSTM | weekly | 33 | 22.67 | 27.64 |
| NO2 | LSTM | daily | 32 | 15.85 | 20.44 |
| NO2 | LSTM | weekly | 32 | 16.40 | 21.21 |
| NOx | LSTM | daily | 33 | 26.08 | 32.84 |
| NOx | LSTM | weekly | 33 | 27.98 | 33.85 |
| Ozone | LSTM | daily | 32 | 5.02 | 5.95 |
| Ozone | LSTM | weekly | 32 | 4.90 | 5.85 |
| PM10 | LSTM | daily | 33 | 70.55 | 91.00 |
| PM10 | LSTM | weekly | 33 | 62.80 | 78.35 |
| PM2.5 | LSTM | daily | 33 | 24.12 | 41.43 |
| PM2.5 | LSTM | weekly | 33 | 18.63 | 33.00 |
| SO2 | LSTM | daily | 33 | 6.42 | 7.95 |
| SO2 | LSTM | weekly | 33 | 7.27 | 8.86 |
| Toluene | LSTM | daily | 32 | 11.31 | 12.03 |
| Toluene | LSTM | weekly | 32 | 10.98 | 12.15 |

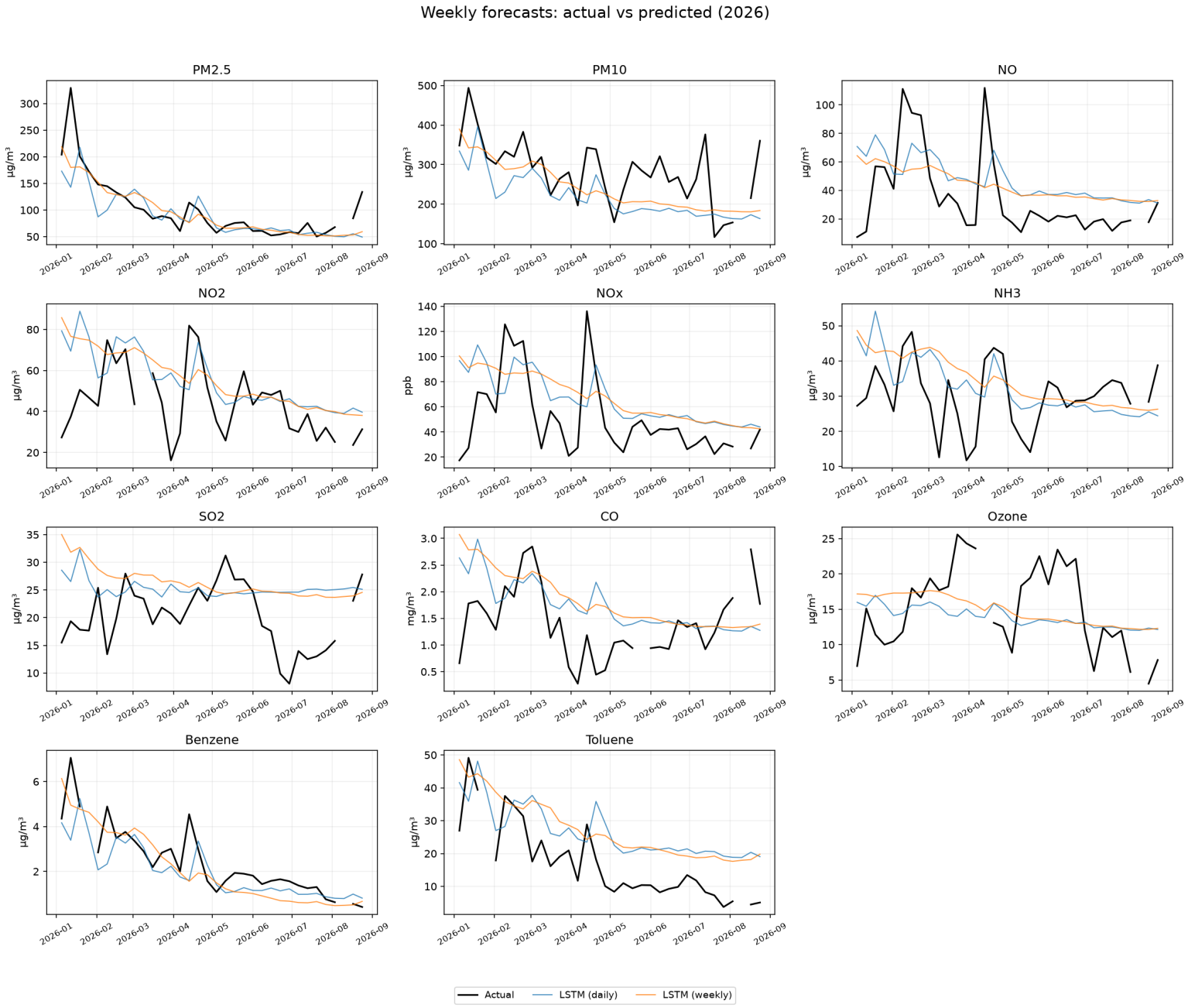
# Weekly MAE comparison



# Weekly RMSE comparison



# Weekly actual and predicted values

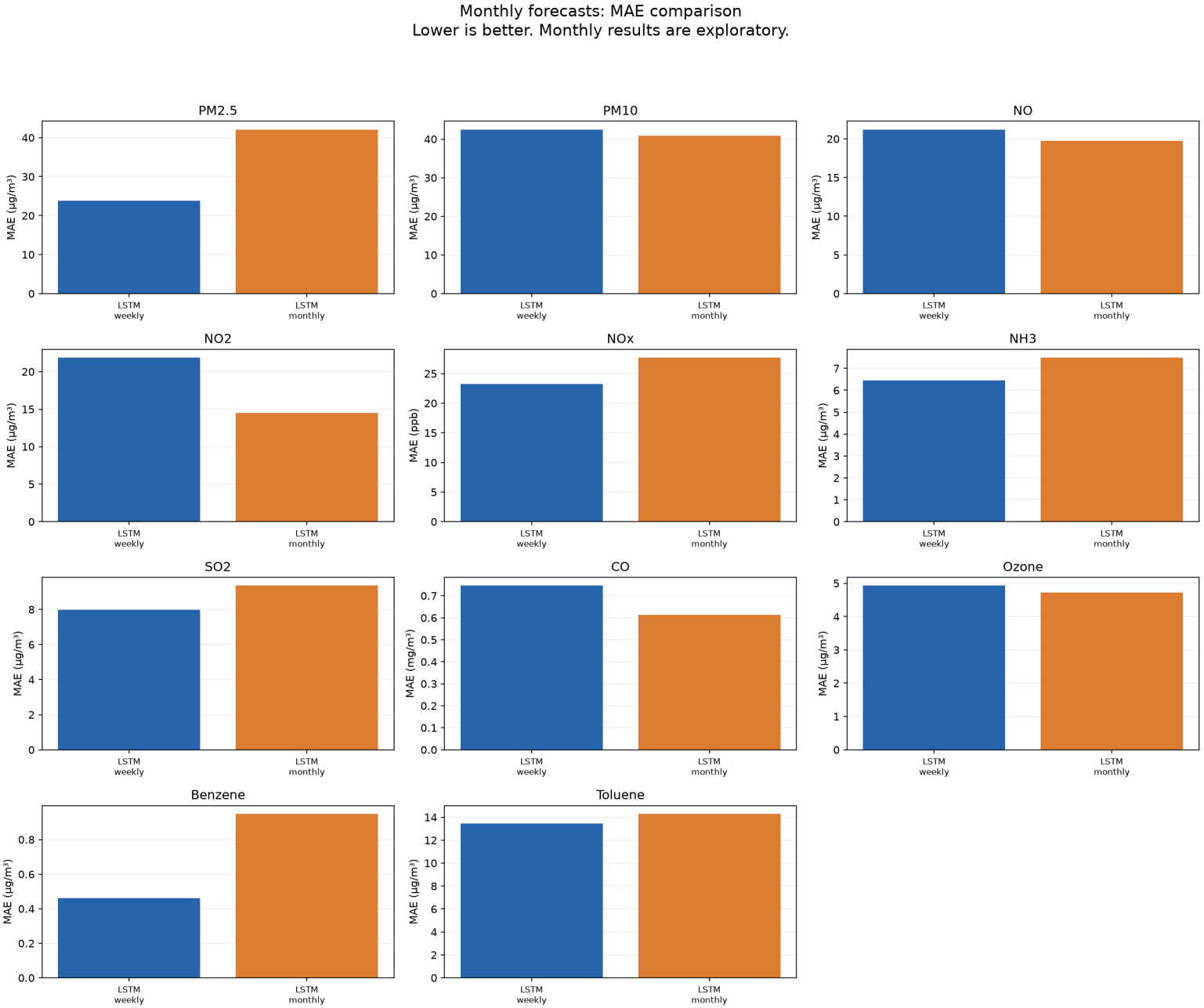


# LSTM monthly errors

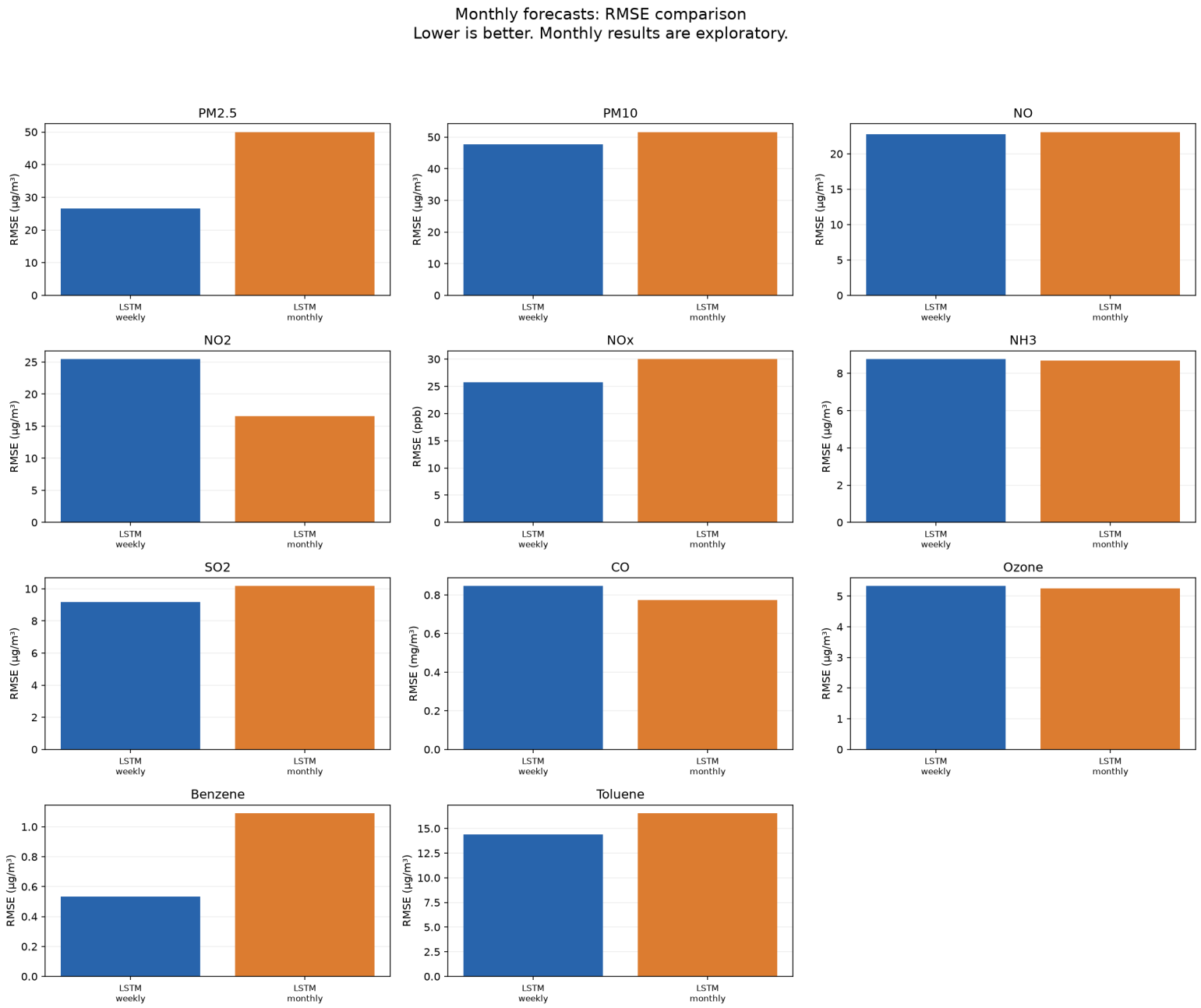
Original concentration units | N is the number of scored periods | Lower is better

| Pollutant | Model | Input scale | N | MAE | RMSE |
|---|---|---|---|---|---|
| Benzene | LSTM | monthly | 8 | 0.95 | 1.09 |
| Benzene | LSTM | weekly | 8 | 0.46 | 0.53 |
| CO | LSTM | monthly | 8 | 0.61 | 0.77 |
| CO | LSTM | weekly | 8 | 0.75 | 0.85 |
| NH3 | LSTM | monthly | 8 | 7.50 | 8.68 |
| NH3 | LSTM | weekly | 8 | 6.45 | 8.76 |
| NO | LSTM | monthly | 8 | 19.74 | 23.08 |
| NO | LSTM | weekly | 8 | 21.21 | 22.78 |
| NO2 | LSTM | monthly | 8 | 14.49 | 16.56 |
| NO2 | LSTM | weekly | 8 | 21.88 | 25.47 |
| NOx | LSTM | monthly | 8 | 27.71 | 30.01 |
| NOx | LSTM | weekly | 8 | 23.24 | 25.75 |
| Ozone | LSTM | monthly | 8 | 4.72 | 5.25 |
| Ozone | LSTM | weekly | 8 | 4.93 | 5.33 |
| PM10 | LSTM | monthly | 8 | 40.88 | 51.59 |
| PM10 | LSTM | weekly | 8 | 42.49 | 47.63 |
| PM2.5 | LSTM | monthly | 8 | 42.10 | 49.99 |
| PM2.5 | LSTM | weekly | 8 | 23.84 | 26.50 |
| SO2 | LSTM | monthly | 8 | 9.36 | 10.17 |
| SO2 | LSTM | weekly | 8 | 7.98 | 9.18 |
| Toluene | LSTM | monthly | 8 | 14.31 | 16.56 |
| Toluene | LSTM | weekly | 8 | 13.47 | 14.38 |

# Monthly MAE comparison



# Monthly RMSE comparison



# Monthly actual and predicted values

